Design and Implementation of a Transduction System for the Measurement of a Second Order Mechanical Oscillator

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OBJECTIVE

This project aims to measure and experimentally determine the frequency response of a second-order oscillatory system with an optical sensor. A system composed of masses, a spring, a motor, and a sensor output frequency can be related to the input frequency/driving frequency from the motor. Key system characteristics such as natural frequency, damping ratio, magnitude ratio, and phase shift can be determined from analysis and are pivotal for understanding this response.

THEORY

Second-Order System

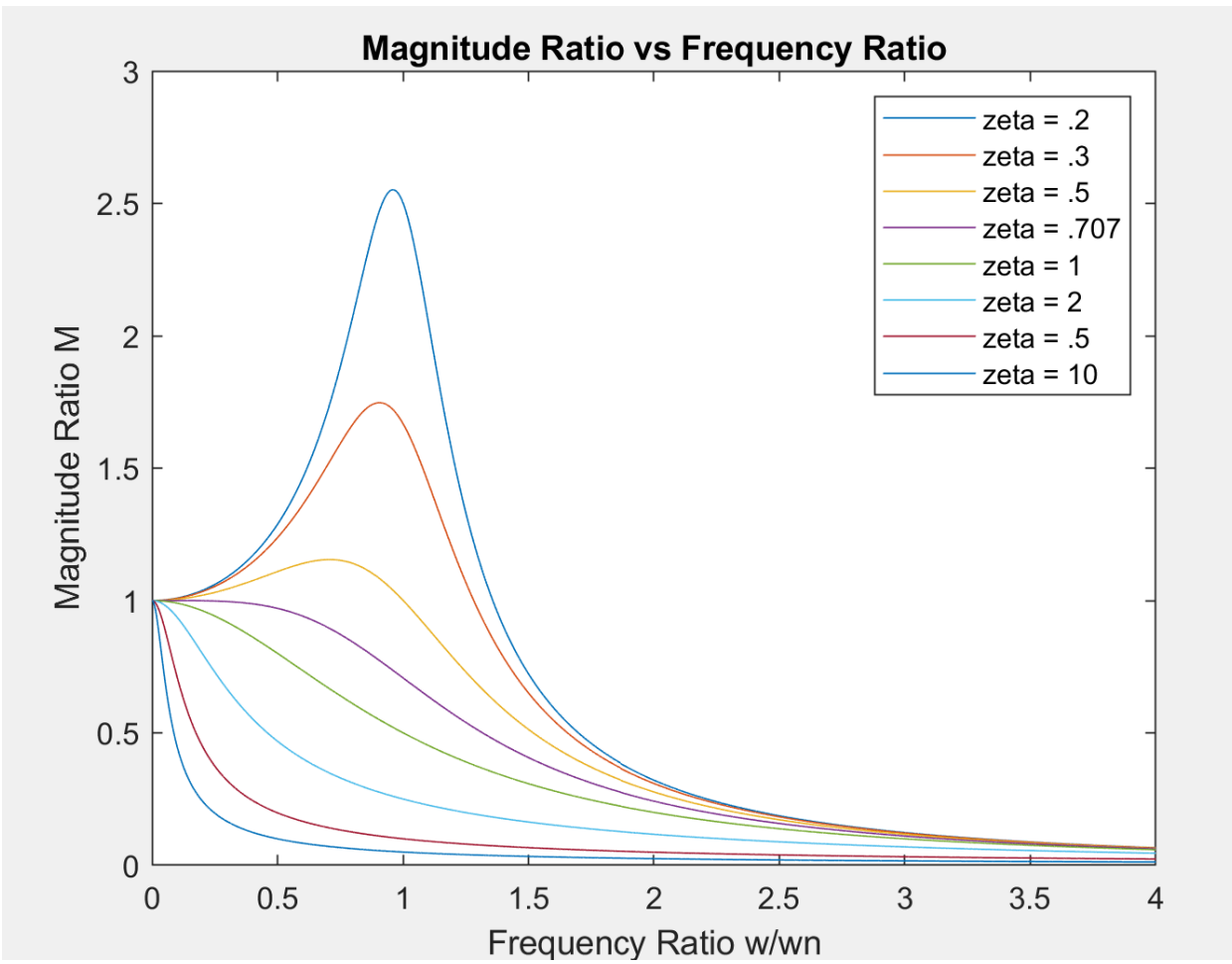

Figure 1. Magnitude Ratio vs. Frequency Ratio Plot

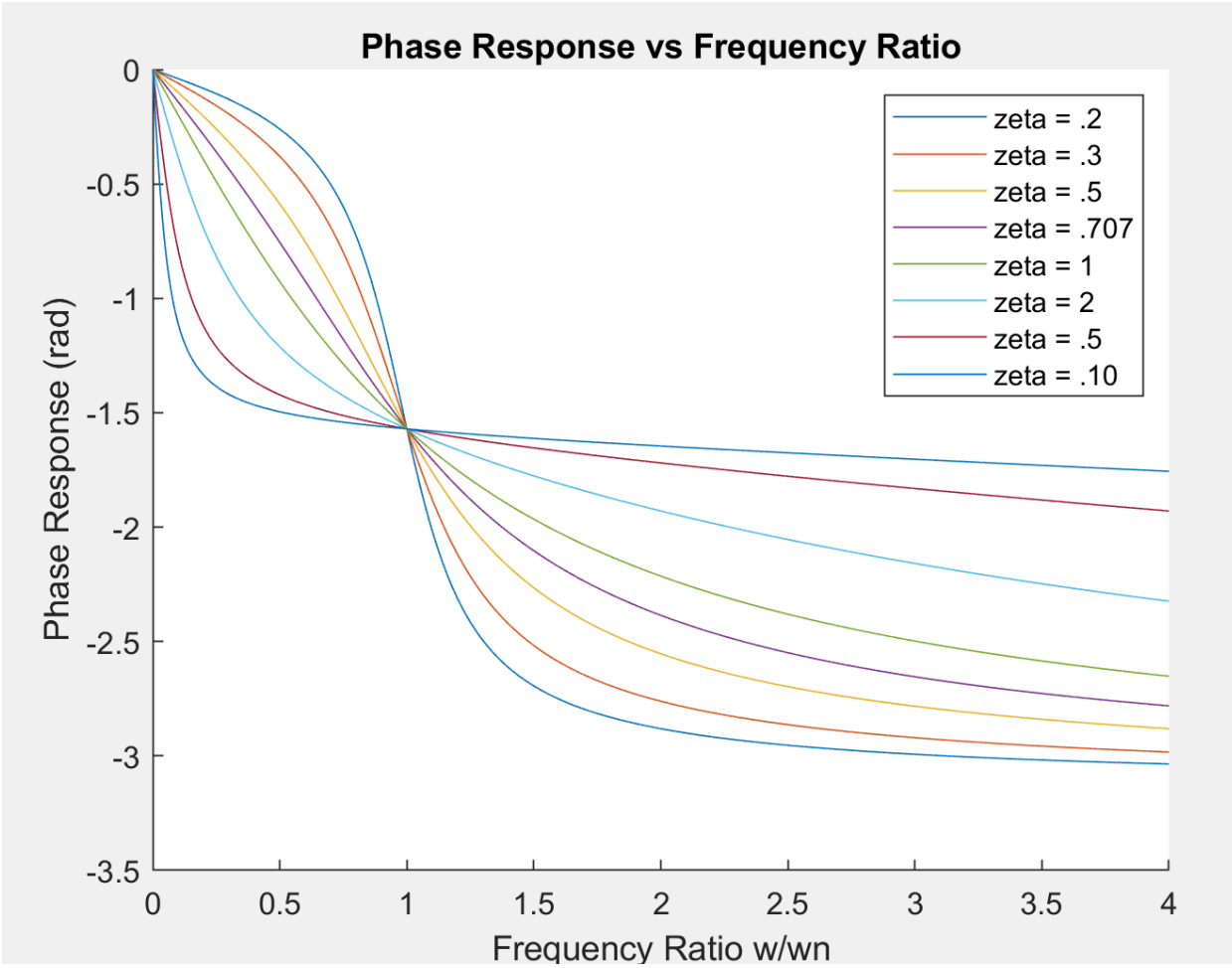

Figure 2. Phase Response vs Frequency Ratio Plot

The basic foundation of the system's operation can be modeled with a spring, and two masses. A bottom mass (m_b) oscillates up and down in a linear motion connected to a top mass (m) via a spring. The bottom mass has a fixed linear displacement which drives the motion of the system and the top mass is allowed to oscillate freely. At low frequencies, both the driving mass frequency and the top mass oscillate with equal displacements/amplitudes but with increasing frequencies, a change in oscillation between the masses also increases.

Equation Summary

While the bottom's mass is unchanging, it can simply be modeled as:

$$x_b = X_b sin(\omega t) \tag{1}$$

where x_b is the mass's function, X_b is its set amplitude, and ω is a varying frequency. On the other hand, the top mass has a more complicated motion as it can move freely. The top mass's oscillations depend on the force due to the spring (f_s) and the force due to the system's damper (f_d) with their respective equations being $F_s =- k(x - x_b)$ and $F_d =- c\dot{x}$ (c equals the damping constant and k the spring constant). Combining all the previous equations while also following Newton's second law ($\sum F_x = m\ddot{x}$), we get an equation of

$$m\ddot{x} + c\dot{x} + kx = KX_b sin(\omega t) \tag{2}$$

After obtaining this equation, we can manipulate it while using the following equations:

$$\omega_n = \sqrt{\frac{k}{m}} \tag{3}$$

$$c_c = 2\sqrt{mk} \tag{4}$$

$$\zeta = \frac{c}{c_c} \tag{5}$$

to describe the ratio of the amplitudes of both masses at a given frequency. Equations (3), (4), and (5) solve for ω_n the natural frequency, c_c the critical damping coefficient, and ζ the damping ratio. This ratio of amplitudes is referred to as the magnitude ratio M(ω). The magnitude ratio relates the magnitude of the output amplitude to the input amplitude as a function of angular frequency in rad/s. M can be calculated in two ways shown below:

$$\frac{X(\omega)}{X_b(\omega)} = M(\omega) = \frac{1}{\sqrt{((1-(\frac{\omega}{\omega_n})^2)^2+(2\zeta\frac{\omega}{\omega_n})^2}} \tag{6}$$

ω represents the varying frequency in the system. The experimental equation for the magnitude ratio is shown with the equation $\frac{X(\omega)}{X_b(\omega)}$ where the ratios from both masses are directly related to each other. This calculation for the magnitude ratio while with given parameters such as the damping ratio and natural frequency, we can obtain a theoretical value for the magnitude ratio using the equation $\frac{1}{\sqrt{((1-(\frac{\omega}{\omega_n})^2)^2+(2\zeta\frac{\omega}{\omega_n})^2}}$.

Using the equations previously we can also find the phase shift/lag. The phase shift represents the delay in response between the driving mass and the freely oscillating mass which can also be found experimentally and theoretically. It can be calculated theoretically by using the equation:

$$\Phi(\omega) = -\tan^{-1}\left(\frac{2\zeta\frac{\omega}{\omega_n}}{1-\left(\frac{\omega}{w_n}\right)^2}\right) \quad (7)$$

The experimental value for phase shift involves two sensors to measure both frequencies of the mass. When graphing the results, one can overlay the frequencies and see the shift firsthand. This value can be calculated simply by finding the difference between the shifts in the data.

Transduction System

For our system, we had an optical proximity sensor called the Sick DT50-P1113. How the sensor captures data is that it emits a light signal toward a given object and measures the time it takes for the light to be reflected. Our devices have a range of 200-800 mm and an analog output between 4mA and 20mA. Using a resistor and Ohm's law, we can convert the amps to a voltage for later calculations.

METHODS & MATERIALS

Optical Sensor Readings

For our experimentation, we were provided with an optical sensor, mounted on a stand and the oscillator system with a disc mounted on the top mass, with white paper taped to it. The taped paper was necessary for the sensor to get a reading, as it did not collect reasonable data from the metal surface of the disc due to the way the sensor measures data by reflecting light. An image of the stand and the oscillator are provided below in Figure 5.

The first step in collecting the data was looking up the specifications of our sensor. From the sensor datasheet, we observed that the output of the sensor was in amps. Our ADC took volts as input signals, therefore we needed to convert our signal. Using the simple relationship of Ohm's law, we decided to connect the output of the sensor and ground using a resistor. The current going through the resistor would result in a voltage that could be collected by ADC. The wiring and block diagram are shown below in Figures 4 and 6.

The output of the sensor was from 4 mA to 20 mA. Considering that we chose to have the data collected using ±5V as our input range for the ADC, we had to use a 250-ohm resistor to observe the signal fully in the input range. However, we could only source two 121-ohm resistors which we placed in sequence and got a 242-ohm resistor which we checked using a multimeter, to confirm that our setup was correct.

One thought that may come here is that the system may clip if the output goes above or below the input range of the ADC. However, we have chosen to use the 200-800 mm range for the sensor. This means that the 4mA output will happen at 200mm measured distance and 20mA would be the output when it measured 800mm. From the height of the sensor on the stand and the highest possible point to which the oscillator could go up, we have found that the system will

not be seeing either 200 or 800 mm as the oscillator only went up to approximately 220mm from the sensor and never below 550 mm. Therefore, we would not experience clipping.

We utilized the MATLAB code provided to us to gather the data. In the code, we made sure that we were using a high sampling frequency so that our Nyquist frequency was above any frequency that the oscillator could output. The oscillator only went up to a maximum of about 5 Hz, however our sampling rate was always higher than 500 Hz. We also made sure that the input range was [-5 5] V.

Our data collection has involved multiple steps beginning with data for calibration. We first took masses from our lab, placed them on the disc surface, and measured the change in height to find the spring stiffness. Next, we collected the data for calibration of the sensor system, by taking measurements of the voltages at various heights and measuring the distance from the sensor to the disc using a ruler. We also recorded the number that the sensor presented on its screen and they always matched the value that was measured with a ruler. This data was used in the weighted linear regression analysis to find the calibration constants. Next, we collected step input data, which allowed us to find the damped natural frequency and damping ratio. We did that by holding the top mass down and releasing it, allowing the system to oscillate until it reached a steady state. We then collected the data for the final analysis of the system to find the experimental values presented in the results which was done by measuring the signal at different driving frequencies. We turned the power knob on the motor system to various power numbers so that we have taken the values across the full power range and more importantly, we went a step further and took the data at smaller increments of power change when we observed that we were close to the resonance frequency. That allowed us to find the resonance frequency of the system more precisely. The last measurements were taken twice both increasing and decreasing the power. This allowed us to analyze the hysteresis of the system.

Finally, to find the phase shift between the bottom and the top mass, we took the second optical light sensor that was available in the lab and we measured the frequency of the bottom mass movement. However, due to the big disk located on the top mass, the second sensor could not be aimed at the bottom mass to measure the frequency directly. Therefore, we constructed a cantilever beam connected to the bottom mass by taping two pencils to the bottom mass and securing a piece of paper and cardboard with duct tape to those pencils shown in Figure 3. The pencils and the paper protrude further than the top disk, which allowed us to aim the second sensor at the paper. However, this method was not the best for the accuracy of the measurements, however, it was the only method we could use considering that we were not allowed to disassemble the system to mount something more rigid to the bottom mass.

All of the data that we collected was double-checked using an oscilloscope connected similarly to the ADC. We compared the wave collected by ADC on MATLAB and the oscilloscope output

at all times to make sure that our wiring stayed intact and that we did not short the circuit or have a bad connection within the system.

We have encountered issues with our system when a connection would get loose, the sensor's measuring range changed, and once we had a resistor fail. The last issue we believe was caused by a faulty resistor. All of those issues were dealt with immediately as we were always carefully observing our data and had multiple spot checks between each sample that we collected.

Figure 3. Image of Dual Sensor Setup with Cantilever Beam attached to the Bottom Mass

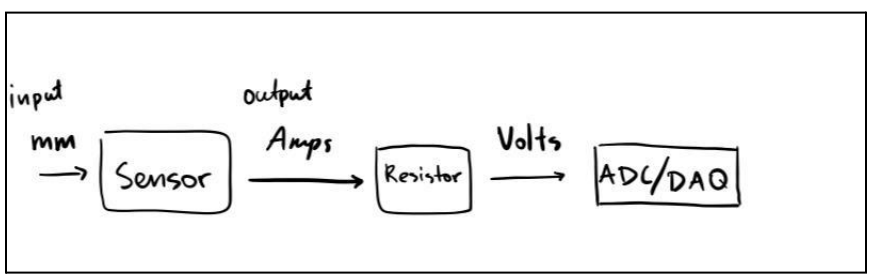

Figure 4. Block Diagram of Optical Sensor Setup

Figure 5. Image of Optical Sensor Setup in the Lab

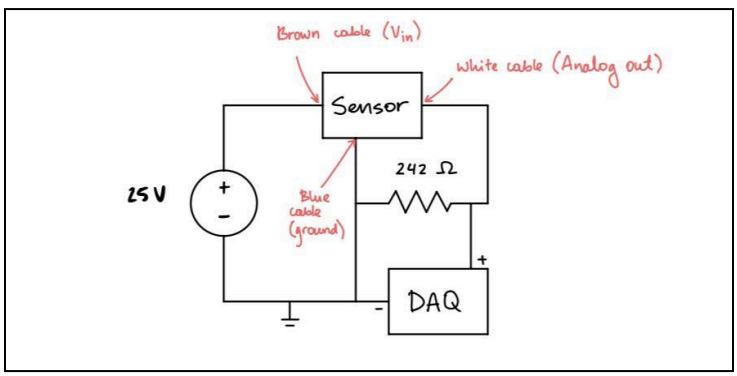

Figure 6. Circuit Diagram of Optical Sensor Setup

DATA / DATA NEEDS

Load [g]	Distance [mm]
0	303
200	330
400	351
600	379
800	397
1000	431
1200	460

1400	473
1600	490
1800	497
2000	512

Table 1. First of Three Spring Constant Calibration Data sets

	Frequency [rad/s]
Oscilloscope	33.11
Stroboscope	33.11

Table 2. Stroboscope and Oscilloscope Values at a 60 Power setting

"Power" from Knob	Corresponding Angular Driving Frequency [rad/s]
20	8.79645943
30	15.07964474
40	22.61946711
50	28.90265241
60	33.92920066
70	33.92920066
80	33.92920066
90	35.18583772
100	35.18583772

Table 3. Driving Frequencies at Varying Knob Power Settings

Equipment List

Instrument	**Brand**	**Model No.**	**Serial No.**	**Resolution**	**Voltage/ Current Range**	**Error Parameters**
Oscillator	-	-	Oscillator Number 5	-	-	-
Oscilloscope	BK Precision	2190E	512G21338	800x480	100-240 V	-
Multimeter	KeySight	34461A	-	10μV-100	DC:	-

				mV	100mV-1000V	
Motor Controller	Minarik Drives	MM232 11C	-	10 bits	115/230 V, 3A	-
Optical Sensor	SICK	DT50-P1 113	-	1 mm	10-30V, 4mA-20 mA	-
Power Supply	Hewlett Packard	E3631A	KR929209 95	-	0-6V, 2.5A 0-±20V, 0.5A	-
DC Gearmotor	Bison	26-999-2 004-027	-	-	90-130V ,0.62A	-
Breakout Box	National Instruments	BNC-20 90	-	-	-	-
DAQ Board	National Instruments	NI-PCI-6221	-	-	-	-
Stroboscope	Extech Instruments	461830	-	±(0.05%rd g + 1 digit)	-	100 - 10000 FPM/RPM
Digital Scale	Triomph	TRKS09	200021-02 302	1 g/0.1 oz	-	Max Weight: 5kg/11 lb
Caliper	Tack Life	DC01	-	0.1 mm	-	-

Table 4. Complete List of All Equipment Used in the Experiment and their respective characteristics

ANALYSIS

Filtering

Given the observed noises within the sinusoidal inputs, a Butterworth low-pass filter was applied to the data. This filter was used to an 6th order with the normalized cut-off frequency of 15 Hz. This allowed for the high noise frequencies to be removed successfully as seen in later plots below.

Optical Sensor Calibration

To calibrate the sensor, we put a weight on the plate, waited until the system settled, and took multiple measurements at that point. This was completed multiple times to obtain enough data for our calibration. To calibrate the sensor, we used a weighted calibration fit for the data. The equations used in this fit include:

$$
\begin{aligned}
a_0 &= \frac{\sum w_i x_i^2 \sum w_i y_{m,i} - \sum w_i x_i \sum w_i x_i y_{m,i}}{B_w} \\
a_1 &= \frac{\sum w_i \sum w_i x_i y_{m,i} - \sum w_i x_i \sum w_i y_{m,i}}{B_w} \\
B_w &= \sum w_i \sum w_i x_i^2 - \left(\sum w_i x_i\right)^2 \\
w_i &= \frac{1}{U_{p,i}^2}
\end{aligned}
\tag{8}
$$

After following these steps, we obtain a relationship:

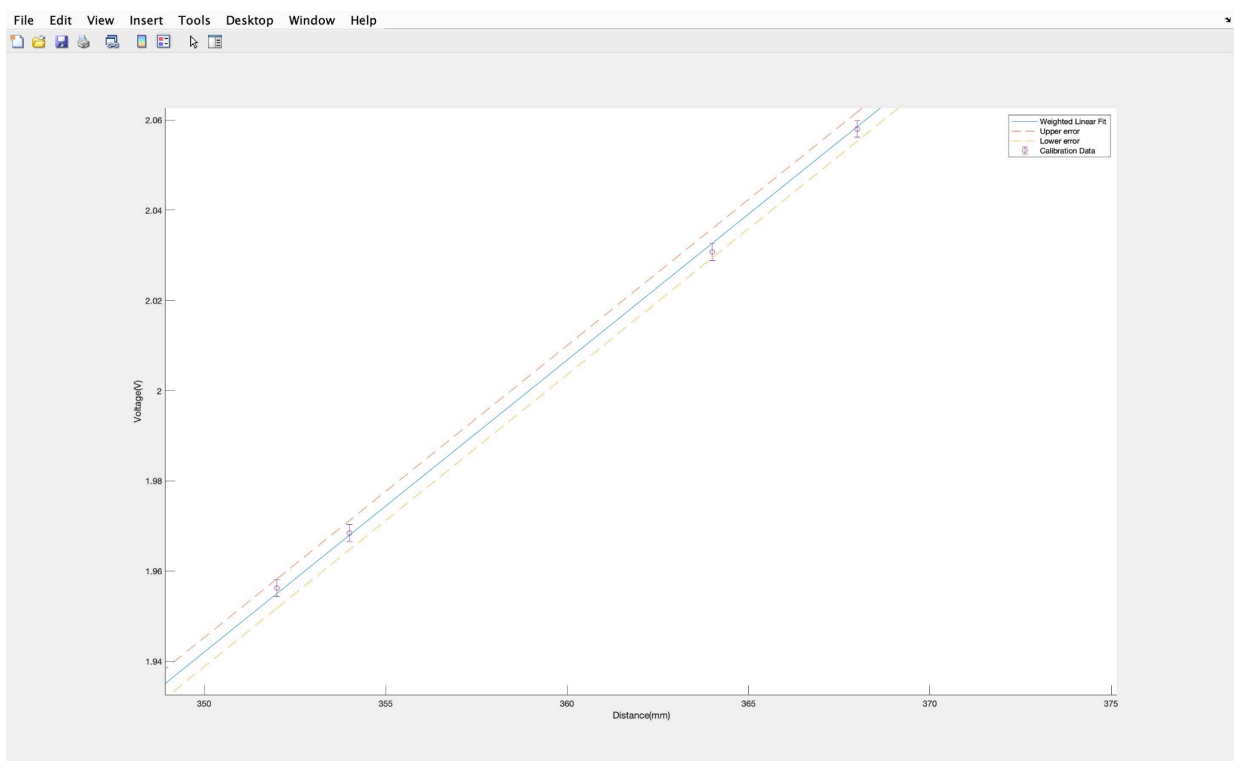

Figure 7. Weighted Calibration Fit for Optical Sensor's Calibration

Using this methodology computationally it was found that

$a_1 = 154.548\ mm/V$

$a_0 = 49.85337\ V$

Where

$$y_{fit,i} = a_1 x_i + a_0$$

Therefore, we find that the sensitivity for the graph is **K = 154.5 mm/V**.

Spring Constant Calibration

To calibrate for the spring constant, we can use a similar approach in which we load weights on the platform and record the distances using the sensor. With weights and distances, this is all that is needed to calculate the spring constant which uses Hooke's Law shown in the equation below:

$$k = -\frac{F}{\Delta x} \tag{9}$$

When applying a weight, we can multiply its mass by the gravitational constant g to obtain the force, then use the distance from the sensor. For example,

$$k = \frac{1\,[kg] * 9.8\,[m/s^2]}{0.431 - 0.303\,[m]}$$
$$k = 76.6\,N/m$$

To calculate a more accurate value, we can plot multiple weights and their distances and obtain the slope between the forces and displacements.

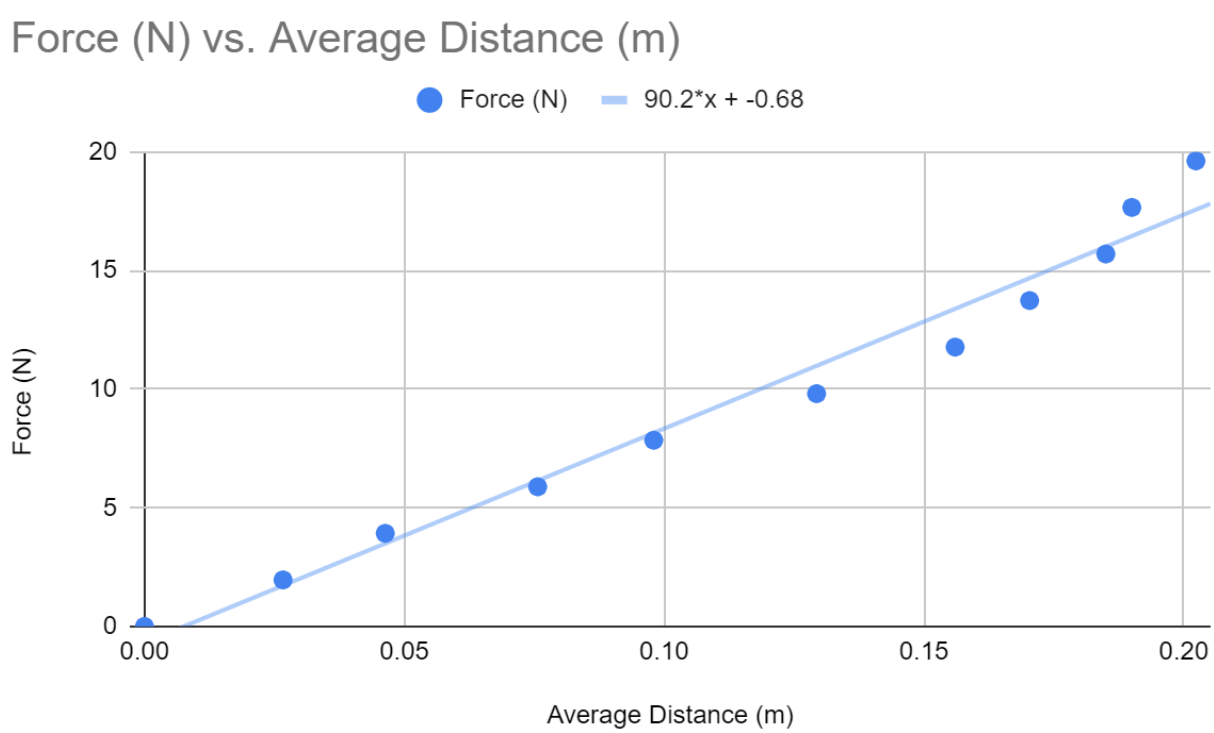

Figure 8. Slope of Varying Forces Due to the Weights on the Spring and their Displacement

The slope indicates our system's spring constant which happens to be **k = 90.2 N/m**.

Natural Frequency

The natural frequency is a crucial system characteristic that is defined as being the frequency that oscillates with no present external forces. We can calculate the natural frequency both theoretically and experimentally. To calculate the natural frequency theoretically, all that is needed is the spring constant and the top mass. The top mass has a value of m = 0.2864 kg and the spring constant we calculated has a value of k = 90.2 N/m. We can solve this using the equation:

$$w_n = \sqrt{\frac{k}{m}} \tag{10}$$

Using the values from above:

$$w_n = \sqrt{\frac{90.2\ N/m}{0.2684\ kg}}$$
$$w_n = 17.72\ rad/s$$

The theoretical value obtained is wn = 2.824 Hz or **wn = 17.72 rad/s**. We can use this value as a basis to compare to the experimentally determined value of natural frequency. To calculate the value experimentally, we first must obtain the damping ratio (ζ) and the resonance frequency (wr). These values are obtained from other sections and are ζ = 0.3946 and wr = 2.4 Hz / 15.08 rad/s. The equation we can use is:

$$w_n = w_r / \sqrt{1 - 2\xi^2} \tag{11}$$

Filling in the values we get:

$$w_n = \frac{15.08\ rad/s}{\sqrt{1-2(0.3946)^2}}$$
$$w_n = 18.17\ rad/s$$

As shown in the previous step, our experimental value for natural frequency is **wn = 17.72 rad/s**.

Damping Ratio

The damping ratio determines how fast the system's amplitude diminishes. Like the natural frequency, the damping ratio (ζ) can also be determined theoretically and experimentally. For the theoretical case, we can use equation (11) and rearrange it to solve for the damping ratio:

$$\zeta = \sqrt{\frac{1}{2} - \frac{w_r^{\,2}}{2\frac{k}{m}}} \tag{12}$$

Plugging in the know values wr = 15.08 rad/s, k = 90.2 N/m, m = 0.2864 kg:

$$\zeta = \sqrt{\frac{1}{2} - \frac{15.08^2}{2\frac{90.2\ N/m}{0.2864}}}$$
$$\zeta = 0.3714$$

Within our system, the theoretical damping ratio is ζ = **0.3714.** Calculating the experimental damping ratio, we perform a step function analysis on specified data. To do this, we first pull down the plate and then release it until it reaches rest while recording the amplitude. This gives us a step function in which we need to find the slope of the envelope that the peaks create in the system. An example of a step function from our experiment demonstrates this idea:

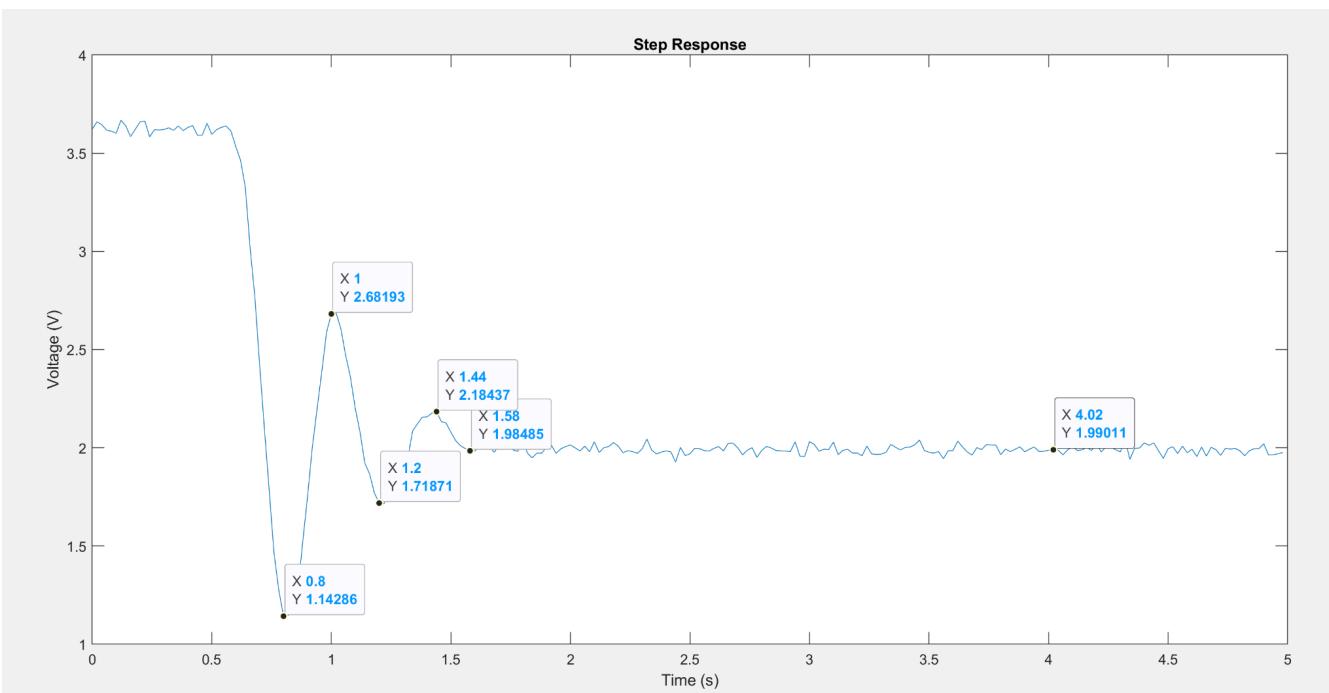

Figure 9. Step Function of Plate Released From Set Position, x-axis is time and y-axis is voltage

Taking from the step function

$y_{inf} = 1.99587\ V$
$y_0 = 1.1428\ V$

We then use the standard equation for finding the damping ratio that involves finding the slope of the descending peaks via the envelope method where y_{env} are the lower peaks of the step function or the normalized top and bottom peaks of the step function.

$$\zeta \cdot \omega_n = ln[\frac{y_{env}(t)-y_\infty}{y_o-y_\infty}]/-t \tag{13}$$

Plotting this relationship yields the following plot, which allows us to find the slope of the linear fit

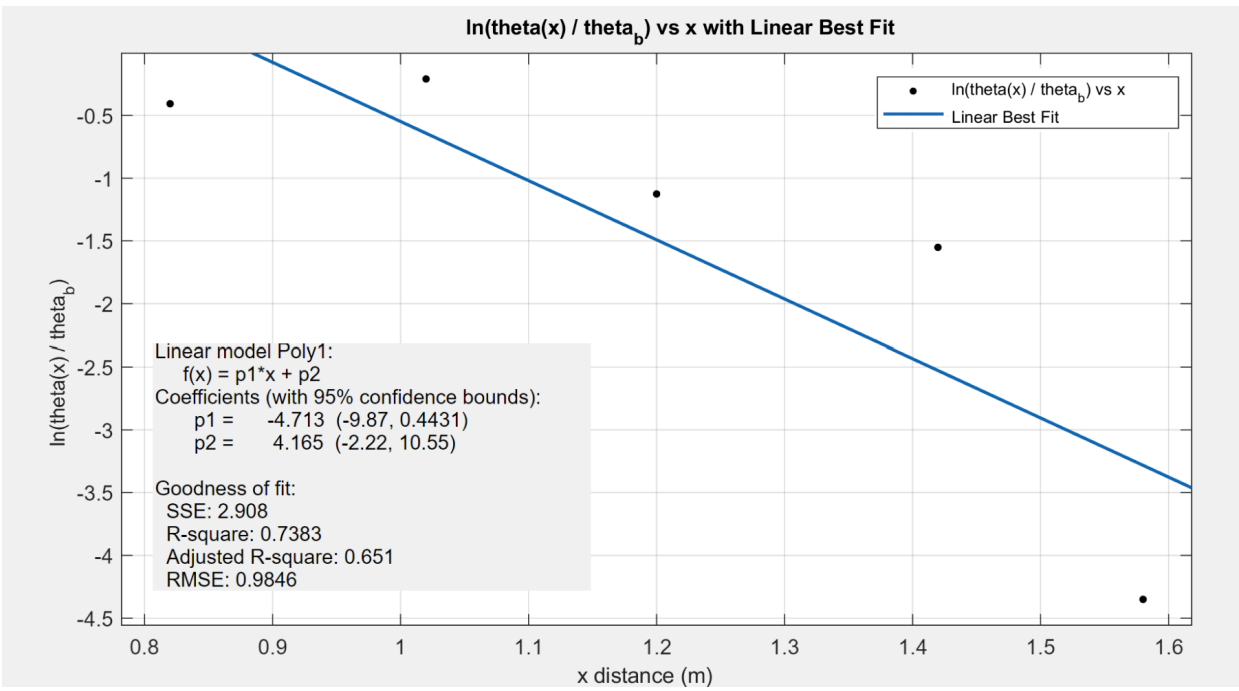

Figure 10. Slope Used to Find Experimental Damping Ratio from Equation (13)

Using the following relationship, we can use the experimental damping frequency along with Equation (13)

$$w_d = w_n\sqrt{1-\zeta^2} \tag{14}$$

The damping frequency can be found via the period of the beginning of the step

$$w_d = \frac{1}{1.71542s-1.14286s} = 1.746541847\ Hz = 1.746541847\ Hz * 2 * pi = 10.9738\ rad/s$$

Given there are now two unknowns with two equations it is possible to calculate the damping ratio experimentally via the step function to be ζ = **0.3946**

Damping Coefficient

The damping coefficient can be described as a coefficient that shows the rate of energy loss in a system until it reaches back to equilibrium. This value is calculated simply by using the equations:

$$c_c = 2\sqrt{\frac{k}{m}} \tag{15}$$

$$\zeta = \frac{c}{c_c} \tag{16}$$

When we combine these equations, we can get the equation for the damping coefficient:

$$c = \zeta * 2\sqrt{km} \tag{17}$$

Plugging in our calculated values for damping ratio, spring constant, and mass:

$$c = 0.3964 * 2\sqrt{90.2\ N/m * 0.2864\ kg}$$
$$c = 4.03\ Ns/m$$

We find that the damping coefficient for the system is **c = 4.03** Ns/m

Linear Distance of Bottom Mass

This step involves measuring the linear motion of the bottom mass. This can be done with simply a caliper by measuring the highest setting of the driving mass and the lowest point relative to the surface below. If you subtract these two the linear distance results in **61.5 mm**.

Driving Frequency

To confirm the driving frequency of the system, we used a stroboscope to match the frequency of the driving mass at a particular setting. How a stroboscope works is that it flashes a light at a certain frequency in flashes per minute. When we use the stroboscope in a dark room facing the driving frequency, the stroboscope is adjusted until the flashes create a still image of the shadow of the driving frequency. More simply, it is adjusted until both the driving frequency and the stroboscope frequency are the same. At the same time, we used an oscilloscope for the frequency of the top mass and we were able to find that the driving frequency and the freely oscillating frequency are identical. At power setting 60 the frequency is **w = 33.05 rad/s** or 5.26 Hz.

Alternatively to the stroboscope, the driving frequency can also be determined via the frequency observed in the system for the top mass given that the frequency will not change from the input to the output. Therefore, the frequency was also observed algorithmically by finding the peaks within the data and the average period within a trial power setting. This allowed for the calculation of frequency from the inverse relationship to period.

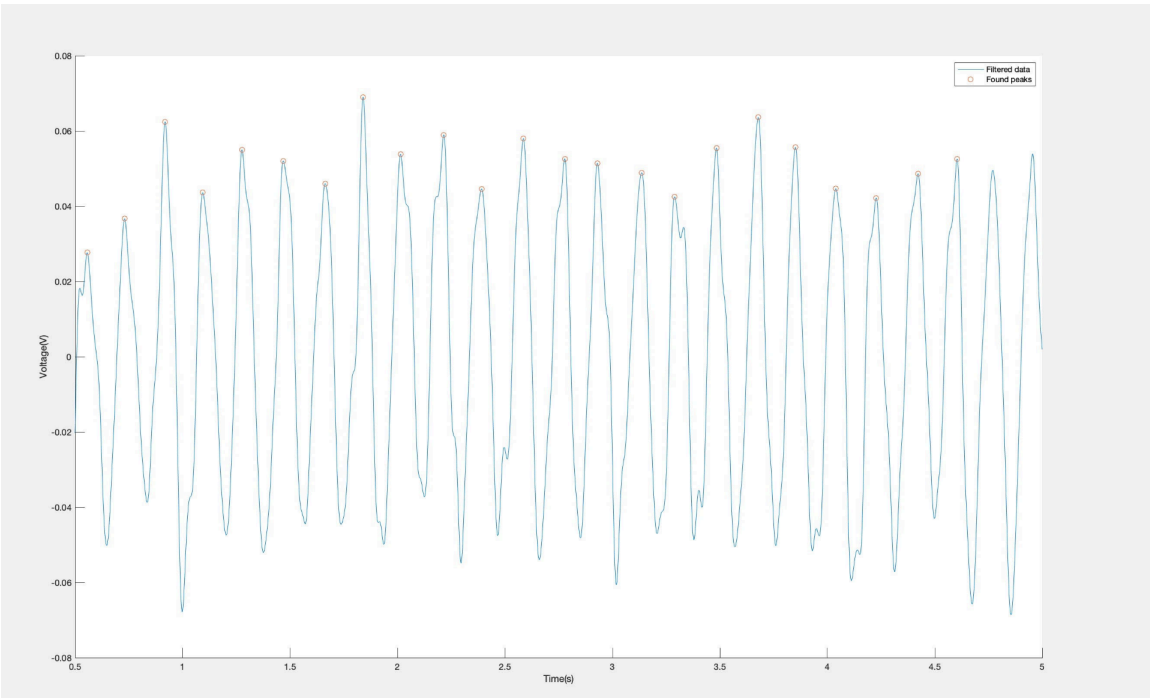

Figure 11. Amplitude vs. Time Graph Displaying the Peak of the System used to Find the Frequency

x(t): Amplitude vs. Time

For our sensor, finding the amplitude is quite simple and only requires a resistor to change the sensor's output from current to voltage. Here is a graph of amplitude vs. time is shown:

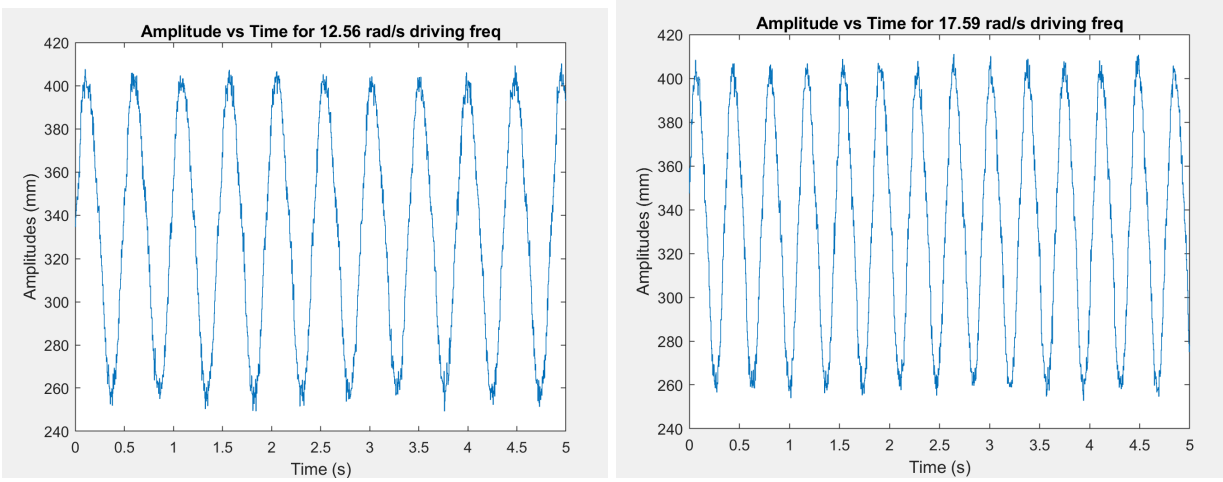

Figure 12. Amplitude vs. Time of when Drive Frequency is at 12.56 rads/s and 17.56 rad/s

It originally plots a voltage vs. time but here we used our sensitivity to convert the voltage to millimeters. At 17.59 rad/s, this is when the amplitude of the system reaches its highest in which we can say that this is our resonance frequency.

Frequency Response

To find the frequency response of our system, we need to collect as many data points for both increasing and decreasing frequency. By doing this, we can visualize and analyze how the system operates throughout the entirety of its function.

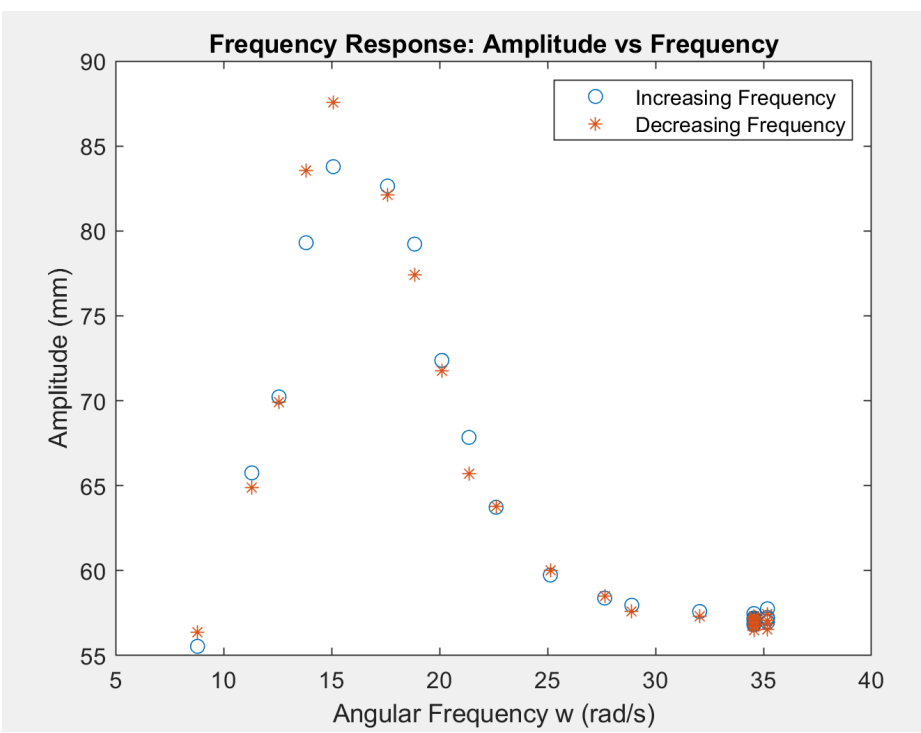

Figure 13. Frequency Response of the System for Both Increasing and Decreasing Data

We can see from this graph that its resonance frequency or the frequency that reaches the highest amplitude is around 15 rad/s. It is also understood that the system reaches a maximum frequency of around 35 rad/s. With this information, we can determine the Nyquist frequency, which determines the minimum frequency for data collection.

$$f_s = 2 * f_{Nyquist} \tag{18}$$

With the max frequency being 35 rad/s or 5.6 Hz:

$$f_s = 2 * 5.6$$
$$f_s = 11.2\ Hz$$

To make sure no aliasing occurs, we need to make sure that the sampling frequency is above **fs = 11.2 Hz**.

Magnitude Ratio

The magnitude Ratio can be found with the following equation

$$M = \frac{X}{X_b} \tag{19}$$

The X value was found as an average of the amplitude at a specific frequency. Xb was measured via manually measuring the maximum amplitude displacement of the bottom mass as this value will not change.

Peaks were found in MATLAB by finding the maximum and minimum values over a set frequency using a FFT. Below is Figure 11. which shows this process of finding the maximum peaks at a sample frequency:

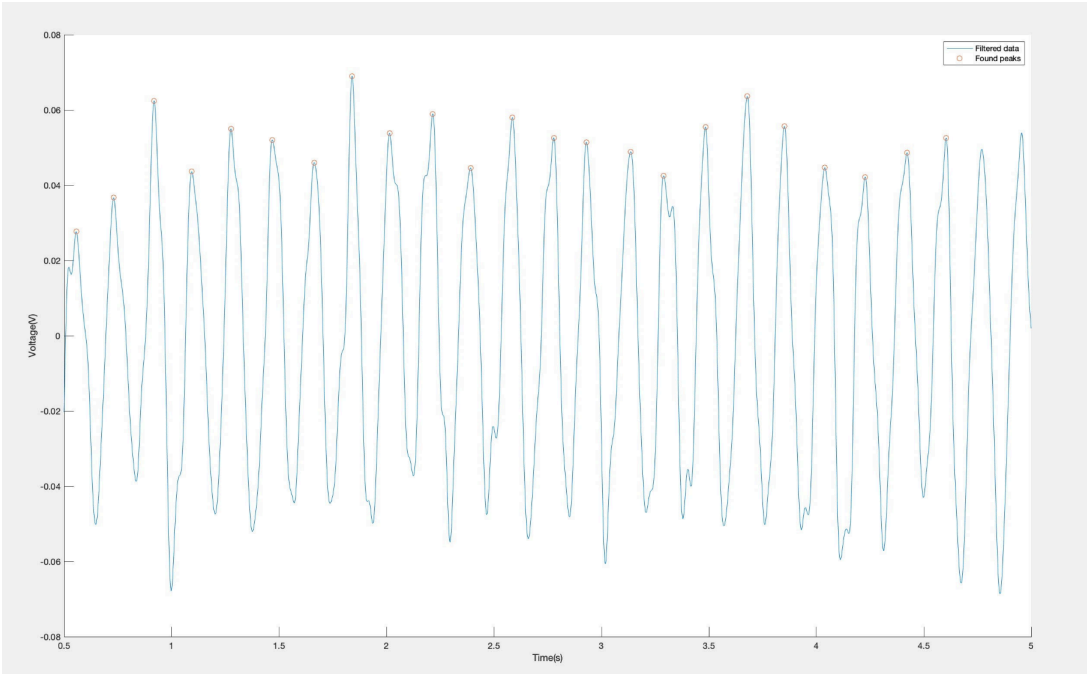

Figure 11. Amplitude vs. Time Graph Displaying the Peak of the System used to Find the Frequency

We can use the equation:

$$X = (Peak_{high} - Peak_{low})/2 \tag{20}$$

and can algorithmically create a plot for X and therefore Magnitude Ratio. Theoretically, the Magnitude Ratio was found via the Equation discussed in the Theory section.

<u>Phase Shift Experimental</u>

As discussed previously, the bottom mass was able to be tracked via a second optical sensor. An example of both of these sensors is shown below. By finding the peaks and times as discussed previously in the same manner, we were able to find the difference between time of the peaks for the first sensor and time of the peaks for the second sensor. These values then allowed us to find the phase offset of the system.

$$\phi = \Delta t_{top-bottom} w_r \tag{21}$$

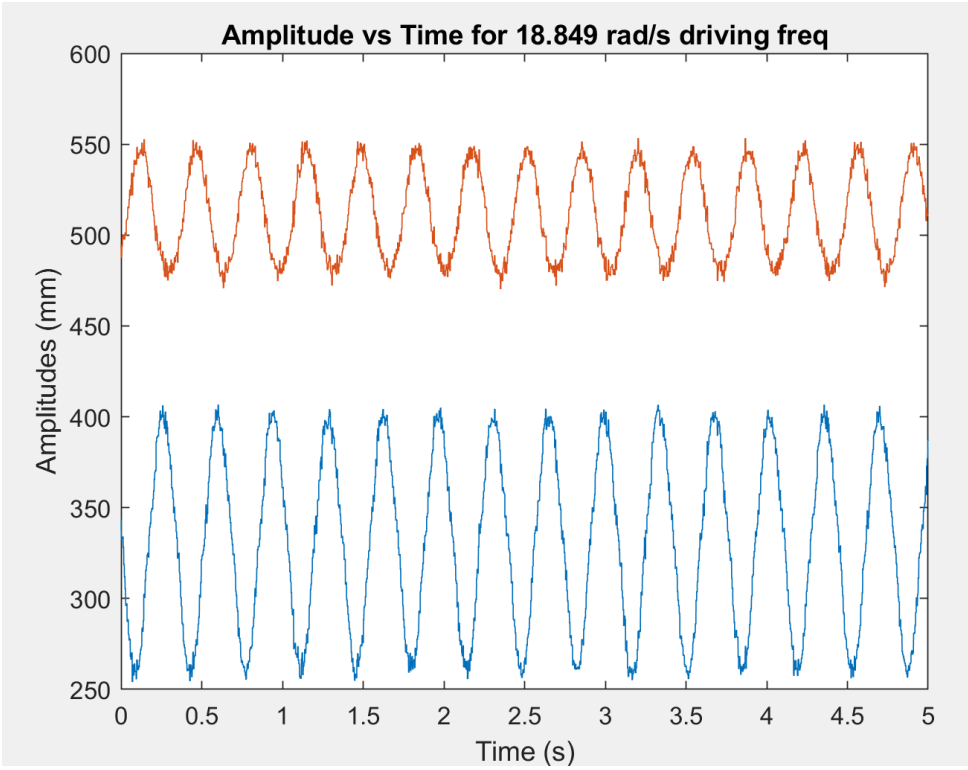

Figure 14. Plot Housing Both Top (Red) and Bottom (Blue) Mass Frequencies Each Using an Independent Optical Sensor

UNCERTAINTY ANALYSIS

Note: Some calculations were done via MATLAB (often in an iterative fashion). Sample calculations are shown for each step to demonstrate the process.

Magnitude Uncertainty

To begin, first the magnitude ratio uncertainty was found. This value given the experimental procedure has both precision and systematic uncertainties and therefore total uncertainty can be expressed via the following equation.

$$U_x = \sqrt{U_p^{\ 2} + U_b^{\ 2}} \tag{22}$$

As seen in the analysis portion of this report, M can be found via the following relationship

$$M = \frac{X}{X_b} \tag{23}$$

First the precision uncertainty can be calculated via the following equation

$$U_p = t_{\nu,95\%}\frac{S_V}{\sqrt{N}} \tag{24}$$

Precision uncertainty can be seen within the amplitude peaks used to create m within the sensor. This can be expressed for both increasing and decreasing datasets as the following.

$$U_{p\ increasing} = 2.042 * (8.8987\ mm / \sqrt{28}) = 3.296\ mm$$
$$U_{p\ decreasing} = 2.042 * (9.3869\ mm / \sqrt{28}) = 3.477\ mm$$

Within U_{Xb} through the experimental procedure there is a measured uncertainty from the caliper used to find this measurement. This can be expressed as the following.

$$U_{Xb} = U_{Caliper\ Res} \tag{25}$$
$$V_{res,\ distance} = .1\ mm$$
$$U_{Res,\ distance} = V_{res}/2 = .05\ mm$$

It is then possible to propagate these errors based upon the partial differential relationship of these two parameters with error as seen in the following.

$$\frac{\delta M}{\delta X} = \frac{1}{\overline{X_b}} \tag{26}$$

$$\frac{\delta M}{\delta X_b} = -\frac{\overline{X}}{\overline{X_b}^2} \tag{27}$$

$$U_{M\ up} = \sqrt{\frac{\delta M}{\delta X}U_x^{\ 2} + \frac{\delta M}{\delta X_b}U_{Xb}^{\ 2}} = \sqrt{\frac{1}{\overline{X_b}}U_x^{\ 2} + -\frac{\overline{X}}{\overline{X_b}^2}U_{Xb}^{\ 2}} \tag{28}$$
$$= \sqrt{\left(\frac{1}{\overline{61.5mm}}3.296\ mm\right)^2 + \left(-\frac{\overline{62.6473mm}}{\overline{61.5}^2\ mm}.05\ m\right)^2} = 0.07393$$

$$U_{M\ down} = \sqrt{\frac{\delta M}{\delta X}U_x^{\ 2} + \frac{\delta M}{\delta X_b}U_{Xb}^{\ 2}} = \sqrt{\frac{1}{\overline{X_b}}U_x^{\ 2} + -\frac{\overline{X}}{\overline{X_b}^2}U_{Xb}^{\ 2}} \tag{29}$$
$$= \sqrt{\left(\frac{1}{\overline{61.5mm}}3.477\ mm\right)^2 + \left(-\frac{\overline{62.6389mm}}{\overline{61.5}^2\ mm}.05\ m\right)^2} = = 0.07609$$

Therefore, we are able to procure uncertainties in this fashion for the the Magnitude Ratio

Spring Constant Uncertainty

The uncertainty of K was then found. This value given the experimental procedure has both precision and systematic uncertainties and therefore total uncertainty can be expressed via the following equation.

$$U_x = \sqrt{U_{p\,fit}^{\;2} + U_b^{\;2}} \tag{30}$$

Because a linear calibration was applied in this situation to for this constant, it is possible to solve for the fit uncertainty

$$U_{p,y,fit} = \pm t_{\nu,95\%} \cdot S_{yx}$$
$$S_{yx} = \sqrt{\frac{\sum (y_{fit,i} - y_{m,i})^2}{\nu}} \tag{31}$$

Where ν is now equal to 6 and $N = 8$ for 8 measurement points from the calibration, thus $\nu = 8 - 2 = 6$

After Applying this to the equation, it can be found that

$$S_{yx} = 1.38\,N$$

Based off the student t table, the t value for 95% confidence where $t_{6,95\%} = 2.447$ as $\nu = 6$,

$$U_{p,\,yfit} = 2.816\,N$$

Acceleration is assumed to be 9.8, mass is assumed to have no uncertainty here as precision weights were used for the calculation. However, the ruler used to measure uncertainties must be accounted for. This can be accomplished as the following.

$$U_{Xb} = U_{Ruler\ Res} \tag{32}$$
$$V_{res,\,distance} = .1\,mm$$
$$U_{Res,\,distance} = V_{res}/2 = .05\,mm$$
$$\frac{\delta k}{\delta l} = -\frac{\bar{F}}{\bar{l}^2} = -\frac{9.8\ N}{(202.3\ mm)^2} \tag{33}$$

Therefore propagating error leads to

$$U_{k\,tot} = \sqrt{\left(\frac{\delta k}{\delta l} U_{B,l}\right)^2 + \left(\frac{1}{a_1} U_{p,yfit}\right)^2} = \sqrt{\left(-\frac{9.8\ N}{(.2023\ mm)^2} * 0.0005\ mm\right)^2 + \left(\frac{1}{90.2\ \frac{N}{m}} 2.816\ N\right)^2}$$

$$U_{k\,tot} = 0.12373\ \frac{N}{m} \tag{34}$$

Mass m Uncertainty

To find mass, a scale was used to measure the top plate attached to the collar. Therefore, the mass has an uncertainty based upon the resolution of the scale.

$$U_{m\,tot} = U_{Scale\ Res} \tag{35}$$

$$U_{Res,\,Scale} = V_{res}/2 = .5\ g$$

$$U_{m\,tot} = .5\ g = .0005\ kg$$

Resonance Frequency Wr Uncertainty

Resonance Frequency uncertainty can primarily be derived by the resolution of steps the data was taken in as this defines how accurate the value can be

$$U_{w_r\,tot} = U_{Scale\ Res} \tag{36}$$

$$V_{res,\,knob} = .2\ Hz = 1.256\ rad/s$$

$$U_{Res,\,knob} = V_{res}/2 = .6283\ rad/s$$

$$U_{w_r\,tot} = .6283\ rad/s$$

Damping Ratio Uncertainty ξ : First Method

To find damping ratio, the uncertainties from m, wr and k were propagated based upon the following equation

$$\xi = \sqrt{-\frac{m w_r^2 - k}{2k}} \tag{37}$$

$$\frac{\delta\xi}{\delta m} = -w_r^2/\left(4k * \sqrt{-\frac{m w_r^2 - k}{2k}}\right) \tag{38}$$

$$\frac{\delta\xi}{\delta m} = -1.256\ rad/s^2 / \left(4 * 90.2\ N/m * \sqrt{-\frac{.2864\ kg * (18.17\ rad/s)^2 - 90.2\ N/m}{2 * 90.2\ N/m}}\right) = -0.00619$$

$$\frac{\delta\xi}{\delta w_r} = -m w_r\ /\left(2k * \sqrt{-\frac{m w_r^2 - k}{2k}}\right) \tag{39}$$

$$\frac{\delta\xi}{\delta w_r} = -.2864\,kg\ *\ 1.256\,rad/s\ /\ (2\ *\ 90.2\,N/m\ *$$

$$\sqrt{-\frac{.2864\,kg\,*(18.17\,rad/s)^2 - 90.2\,N/m}{2\,*\,90.2\,N/m}}\ )\ = -\ 0.00282$$

$$\frac{\delta\xi}{\delta k} = mw_r^{\ 2}/(4k^2 * \sqrt{-\frac{mw_r^{\ 2}-k}{2k}}\ ) \tag{40}$$

$$\frac{\delta\xi}{\delta k} = .2864\,kg\ *\ 1.256\,rad/s^2\ /\ (k\ *\ (90.2\,N/m)^2\ *$$

$$\sqrt{-\frac{.2864\,kg\,*(18.17\,rad/s)^2 - 90.2\,N/m}{2\,*\,90.2\,N/m}}\ )\ = -\ 0.00001$$

$$U_{\xi\,tot\,1} = \sqrt{(\frac{\delta\xi}{\delta m}U_m)^2 + (\frac{\delta\xi}{\delta w_r}U_{w_r})^2 + (\frac{\delta\xi}{\delta k}U_m)^2} = 0.00177 \tag{41}$$

<u>Damping Ratio Uncertainty ξ : Second Method</u>

The second method of calculating the damping ratio is related to the step function envelope methodology. Given this, there are numerous sources of uncertainty present that cannot be uncoupled efficiently. Several major considerations can be analyzed concerning the methodology of calculation.

Considering that ξ was solved based upon the following set of coupled equations

$$\xi = \sqrt{\frac{w_r^{\ 2} - w_n^{\ 2}}{w_n^{\ 2}}} \tag{42}$$

$$\xi = ln(\frac{y_{enc} - y_{inf}}{y_0 - y_{inf}})\ /\ -\ t\ *\ w_n \tag{43}$$

The one of most dominant uncertainties will be related to w_r. Therefore The total uncertainty of ξ is approximately equal to the uncertainty propagated in natural frequency,

$$\frac{\delta\xi}{\delta w_r} = \frac{w_r}{w_n\sqrt{w_r^{\ 2} - w_n^{\ 2}}} = 0.25515 \tag{44}$$

$$U_{\xi\,tot\,2} \approx \sqrt{\frac{\delta\xi}{\delta w_r}(.6283\,rad/s)} \approx 0.16031 \tag{45}$$

Natural Frequency Wn Uncertainty: First Method

For the first method to solve natural frequency, the uncertainties of k and m are already known so this can be directly propagated

$$w_n = \sqrt{\frac{k}{m}} \tag{46}$$

$$\frac{\delta w_n}{\delta k} = \frac{1}{2\sqrt{k}\sqrt{m}} \tag{47}$$

$$\frac{\delta w_n}{\delta m} = -\frac{\sqrt{k}}{2m^{3/2}} \tag{48}$$

$$U_{tot\, w_n 1} = \sqrt{\left(\frac{\delta wn}{\delta k} U_{k\, tot}\right)^2 + \left(\frac{\delta wn}{\delta m} U_{m\, tot}\right)^2} = \sqrt{\left(\frac{1}{2\sqrt{90.2}\sqrt{0.2864}} \cdot 0.12373\right)^2 + \left(\frac{\sqrt{90.2}}{2 \cdot 0.2864^{3/2}} \cdot 0.0005\right)^2}$$

$$U_{tot\, w_n 1} = 0.01970\, rad/s \tag{49}$$

Natural Frequency Wn Uncertainty: Second Method

For the second method to solve natural frequency, the uncertainties of k and m are already known so this can be directly propagated according to the following equational relationship

$$w_n = w_r / \sqrt{1 - 2\xi^2} \tag{50}$$

$$\frac{\delta w_n}{\delta w_r} = 1/\sqrt{1 - 2\xi^2} = 1/\left(\sqrt{1 - 2(.37135)^2}\right) = 1.17509 \tag{51}$$

$$\frac{\delta w_n}{\delta \xi} = \frac{2 w_r \xi}{(1-2\xi^2)^{3/2}} = 103.3971 \tag{52}$$

$$U_{tot\, w_n 2} = \sqrt{\left(\frac{\delta w_n}{\delta w_r} U_{w_r}\right)^2 + \left(\frac{\delta w_n}{\delta \xi} U_{\xi}\right)^2} = 0.760\, rad/s \tag{53}$$

Damping Coefficient Uncertainty

To solve damping coefficient uncertainty, the uncertainties of zeta, k and m are already known so this can be directly propagated according to the following equational relationship

$$c = \xi * 2\sqrt{km} \tag{54}$$

$$\frac{\delta c}{\delta \xi} = 2\sqrt{km} = 2\sqrt{90.2\, N/m * .2864\, kg} = 10.165 \tag{55}$$

$$\frac{\delta c}{\delta k} = \frac{\xi\sqrt{m}}{\sqrt{k}} = (.37135 * \sqrt{.2864\ kg})/\sqrt{90.2\ N/m}) = 0.02092 \quad (56)$$

$$\frac{\delta c}{\delta m} = \frac{\xi\sqrt{k}}{\sqrt{m}} = (.37135 * \sqrt{90.2\ N/m})/\sqrt{.2864\ kg}) = 6.59022 \quad (57)$$

$$U_{\xi\, tot} = \sqrt{(\frac{\delta\xi}{\delta m}U_m)^2 + (\frac{\delta\xi}{\delta w_r}U_{w_r})^2 + (\frac{\delta\xi}{\delta k}U_m)^2} = 0.01846\ Ns/m \quad (58)$$

Phase Shift Uncertainty

To solve for phase shift uncertainty, U precision can be calculated via the time differences between top mass peaks and bottom mass peaks. It is important to note that for different frequencies the number of time differences changes and therefore Up at different frequencies due to a different number of total oscillations recorded. Therefore only sample calculations are provided.

For a driving frequency $33.9292\ rad/s$ it was observed that there were 24 cycles in the time lag was found with

$$U_p = t_{\nu,95\%}\frac{S_V}{\sqrt{N}}$$

$$U_{\phi\, p\ @\ 33.9292\ rad/s} = 2.064 * (0.3322\ rad/\sqrt{24}) = 0.1400 rad \quad (59)$$

Using the relationship found to find phase shift, there is also an observed propagated error from the resonance frequency

$$\phi = \Delta t_{top-bottom}\, w_r \quad (60)$$

$$\frac{\delta\phi}{\delta w_r} = \overline{\Delta t_{top-bottom}} \quad (61)$$

$$U_{\phi\, tot} = \sqrt{U_{\phi\, p}^{\ 2} + \frac{\delta M}{\delta X_b}U_{\phi\, b}^{\ 2}} = \sqrt{U_{\phi\, p}^{\ 2} + \overline{\Delta t_{top-bottom}}U_{w_r}^{\ 2}} \quad (62)$$

$$U_{\phi\, tot} = \sqrt{0.1400^2\ rad + 0.0975\ s * .6283\ rad/s} = 0.41642\ rad \quad (63)$$

RESULTS

The Results of two examples of the time-dependent position of the top collar as measured by the sensor within the analyzed range. It is important to note that this data is shown before filtering but with the calibration applied. These values are what were expected as the physical distances match what was empirically observed. This was observed for trials going up and going down with similar results. There are some larger spikes seen in the data from different noise sources. However, filtering this data using a low pass filter removes these noise sources.

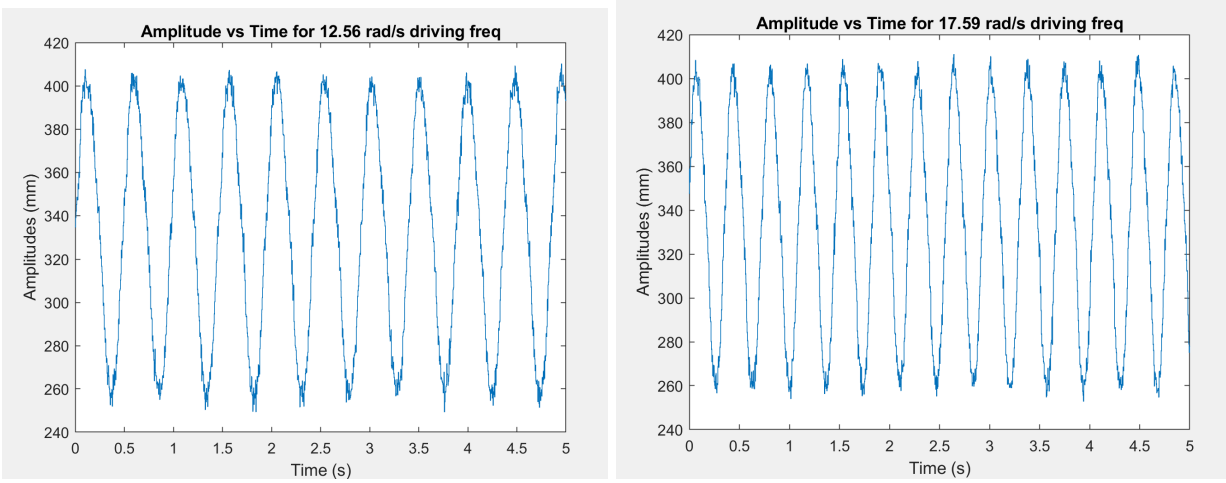

Figure 15. Time-Dependent Position of Top Collar Over Time Samples

The Results of the Experimental vs Theoretical Magnitude Ratio can be seen below. It is clear that both the Experimental and Theoretical ratio approximately follow the same expected curve. The maximum M is seen at approximately the resonance frequency as expected and then Magnitude Ratio then decreases before and after that point. It was interesting to visualize that the Magnitude Ratio appeared to reach a value of one at higher frequencies in experimental cases while the theoretical M became smaller than 1. This could be errors within the experimental set up given that there was paper that vibrated significantly on top of the plate at higher frequencies. This correctly matched the theoretical chart found in the theory section as well given the damping ratio and driving frequencies calculated.

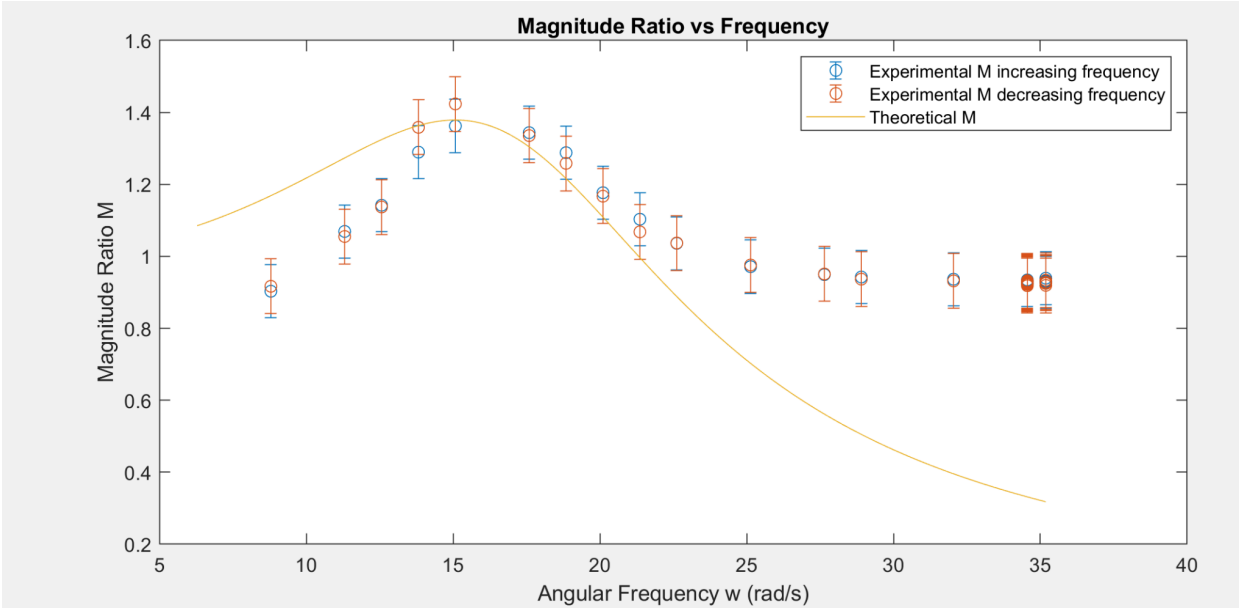

Figure 16. Magnitude Ratio of Experimental and Theoretical Trends

Likewise, the Results between Phase Response vs Frequency for both Experimental and Theoretical approximately matched with one another. As we developed a methodology using a piece of paper and pencil cantilever beam to measure with the second sensor for the bottom, it is clear that there will be errors induced through the additional vibration from the beam on top of what is expected from the second mass. This is clear in the data as at smaller frequencies, phase shift matches much closer to the theoretical than at larger frequencies where the beam most likely hits its own resonance frequency and increases the amplitude of its vibration, leading to more errors in experimental results. This correctly matched the theoretical chart found in the theory section as well given the damping ratio and driving frequencies calculated.

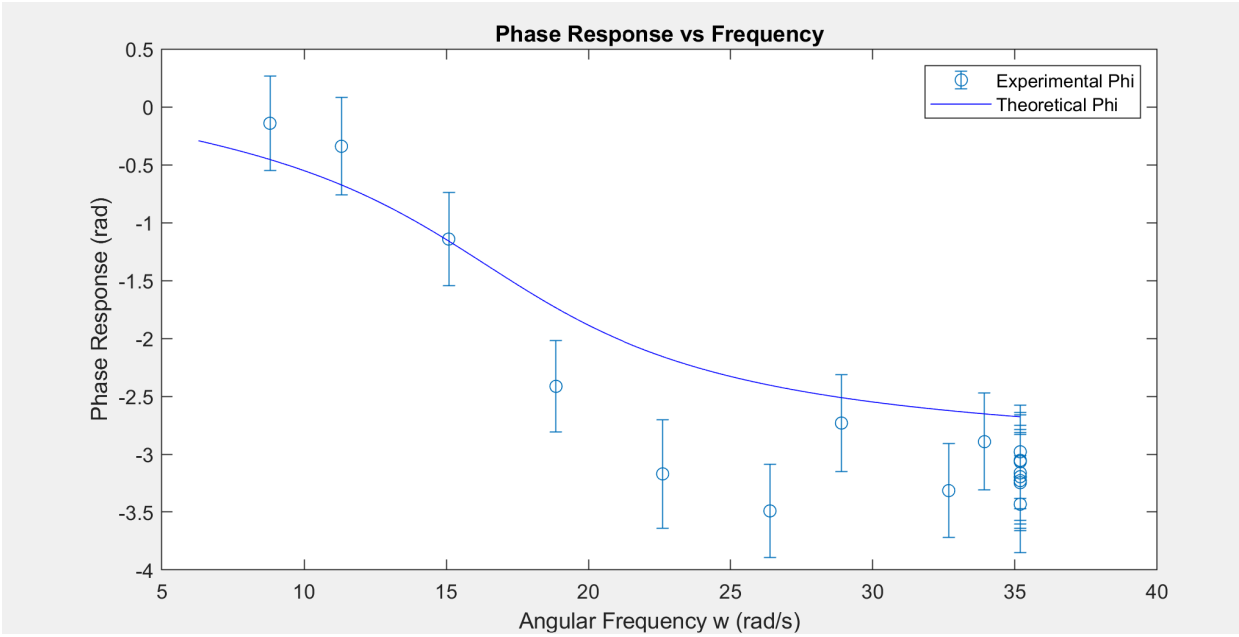

Figure 17. Phase Lag of Experimental and Theoretical Trends

Overall, all solved System Characteristics and Uncertainty were approximately what was expected of the system. Uncertainties for all situations were reasonable and made sense with respect to the data. In cases where multiple methodologies were used to solve for the same value. The values were always close to one another, further confirming the results of this procedure.

$k\ [N/m]$	$w_n\ [rad/s]$	ξ	$c\ [Ns/m]$	$m\ [kg]$
$90.2 \pm .12$	$17.7 \pm .020$	$.373 \pm .0018$	$4.03 \pm .018$	$.286 \pm .0005$
	$18.2 \pm .76$	$.395 \pm .16$		

Table 5: System Characteristics & Uncertainty

DISCUSSION/CONCLUSION

Overall, after careful collection and analysis, we were quite satisfied with our results as our experimental data matched our theoretical calculations with a low margin of uncertainty for all calculated values. As seen from the theory section, in pure theory, the observed trends correctly mirrored the expected trends for the given driving frequency and damping ratio calculated.

Despite the successes observed within this experiment, there were some areas where experimental results were slightly different than theoretical results. This was seen clearly within the experimental phase lag and comparison to theoretical values. To measure the phase lag we assumed that there was no phase shift between the self-made cantilever beam and the bottom mass. However in reality, when we hit the resonant frequency of the beam there was a further

phase shift between the paper beam and the bottom mass. This is evident within the data as there is an increase of phase lag at higher frequencies than expected but at smaller frequencies, the experimental and theoretical match nearly perfectly. Therefore, there is an observed experimental procedure error leading to this trend.

Likewise, the Magnitude Ratio was also observed to have slight deviations from theory. The experimental Magnitude Ratio was observed to correctly follow the same trend as theoretical and have approximately the same magnitude. At resonance, it was clear that both plots peaked at the same value as expected for this analysis. As described in the results, M appeared to reach a value of one at higher frequencies in experimental cases while the theoretical M became smaller than one. This again could be attributed to systemic errors within the experimental setup given that there was paper that vibrated significantly on top of the plate at higher frequencies.

Other discrepancies can be attributed to this system not being a true 2nd-order linear system. There is inherent nonlinearity within the sensors and system parameters used that were unaccounted for as seen via sensor response. This means that there will be deviations from theoretical values even with the most precise and accurate measurements. Another source of error comes from the frictional forces and drag forces present within the system. It was clear that within this homemade system, there were high levels of friction between the collar and the bar, which could have led to further deviations unaccounted for in theory.

If given the opportunity we would make several improvements to the experimental procedure to achieve even more accurate and precise results. First of all, we would redesign the cantilever beam by actually disassembling the instrument to a material with a very high resonant frequency. In this experiment, we were not allowed to take off the bottom collar, but we believe that modifying the system would lead to better results. Lubricating the areas of friction seen between the collar and the pole would lead to a further reduction in frictional effects. Lastly, by using a nonlinear estimation and analytical technique, this would be able to better match the data, oftentimes this can be used via non-linear least squares techniques as well as extended estimation filtering via sensor fusion techniques.

Ultimately, the project's goal of measuring and experimentally figuring out a second-order oscillator system's frequency response was accomplished. We established a relationship between the motor's input frequency and the system response by dissecting a system that included masses, a spring, a motor, and a sensor output frequency. Important characteristics such as the spring constant, damping ratio, natural frequency, phase shift, and magnitude ratio were determined through thorough analysis. These results provide the foundation for additional research and optimization in related fields and are essential for an adequate understanding of the response dynamics of the system.

Appendix

Appendix A. Complete Table of Driving Frequencies at Varying Knob Power Settings

‘Power' from Knob	Corresponding Angular Driving Frequency [rad/s]
20	8.79645943
24	11.30973355
26	12.56637061
28	13.82300768
30	15.07964474
32	17.59291886
34	18.84955592
36	20.10619298
38	21.36283004
40	22.61946711
44	25.13274123
48	27.64601535
50	28.90265241
54	31.41592654
58	33.92920066
60	33.92920066
64	33.92920066
68	33.92920066
70	33.92920066
74	33.92920066
78	33.92920066
80	33.92920066
84	33.92920066
88	33.92920066
90	35.18583772
94	35.18583772
98	35.18583772
100	35.18583772